

AMENDMENT NO. 1 AUGUST 2020

TO

**IS 9168 : 2018 AUTOMOTIVE VEHICLE —
ALL RUBBER FLAPS FOR PNEUMATIC TYRES —
SPECIFICATION**

(Second Revision)

(Second cover page, Foreword, Para second, Line 1) — Substitute ‘1996’ for ‘1991’.

(Page 2, clause 5.1.2) — Substitute the following for the existing clause:

‘5.1.2 Flap with Overlap Joint

Where the joint is visible in a finished flap, the adhesion strength at the joint shall be checked and shall be not less than 1.0 MPa, when tested in accordance with the test procedure outlined in Annex B.’

(Page 2, clause 5.2) — Substitute the following for the existing clause:

‘5.2 Hardness

The shore hardness of the unused finished flap before heat ageing shall lie between 45 and 65, when measured by Shore Type ‘A’ Durometer. The hardness shall be measured at or on test piece cut from centre of the flap as per test method given in IS 3400 (Part 2).’

(Page 2, clause 5.3) — Substitute the following for the existing clause:

‘5.3 Retention of Properties on Ageing

The test piece having ‘Type 1 dumb bell shape’ shall be prepared from the flap according to IS 3400 (Part 1). Dumb bells of Type 1 punched directly from the flap from a portion having best possible flatness may also be used for testing. The dumb bells shall be subjected to ageing in an air oven at $100^{\circ}\text{C} \pm 2^{\circ}\text{C}$ for 24 h

according to IS 3400 (Part 4). The tensile strength and elongation before and after ageing shall as given below:

- a) *Before heat ageing:*
 - 1) Hardness, shore A: 45 to 65
 - 2) Tensile strength, kg/cm² : 90 Min
 - 3) Elongation at break, percent: Min 300
- b) *After heat ageing at 100°C for 24h (shall not vary more than):*
 - 1) Change in hardness shore A: -5 to 10 units
 - 2) Tensile strength, kg/cm²: + 15 percent to -50 percent
 - 3) Elongation at break: + 15 percent to - 50 percent

Example:

If the original elongation at break percent is 400, the aged elongation should not vary more than (+) 60 to (-) 200.'

(Page 2, clause 7) — Substitute the following for the existing clause:

‘7 BIS CERTIFICATION MARKING

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act*, 2016 and the Rules and Regulations framed thereunder, and the products may be marked with the Standard Mark.'

(Page 3, clause **B-3.2**) — Substitute the following for the existing clause:

‘B-3.2 Testing shall be carried out on three test pieces of which at least two pieces shall be taken from the joint. Mean joint adhesion strength of the three test pieces shall be reported for compliance.'

(Page 4, clause **B-3.3**) — Substitute the following for the existing clause:

‘B-3.3 The joint adhesion strength which is indicated in **5.1.2** shall be calculated from the formula:

$$\text{Joint adhesion strength} = \frac{L}{A} \text{ MPa}$$

where,

L = breaking load of the dumb-bell strip, in N; and

A = cross sectional area of the unstrained test piece, in mm^2 .’

(Page 4, *clause C-1.2*) — Substitute the following for the existing clause:

‘**C-1.2** Month code circle and year to be engraved not more than 200 mm away from the valve hole.’